WOXSEN UNIVERSITY • INDUSTRY-INTEGRATED TECHNICAL COMMUNICATION PROJECT

The Impact of Generative AI on Cognitive Offloading: Assessing Information Retention and Research Depth Among University Undergraduates

LEAD RESEARCHER	ACADEMIC PROGRAM	SAMPLE SIZE	PUBLICATION DATE
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1. Executive Summary & Abstract

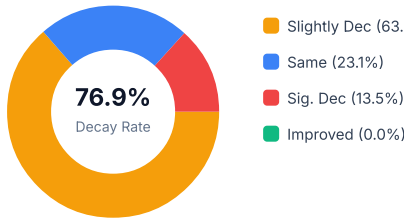
The integration of Generative Artificial Intelligence (GenAI)—including Large Language Models (LLMs) such as ChatGPT, GitHub Copilot, and Gemini—has radically altered how university undergraduates retrieve, process, and consolidate academic knowledge. While LLMs offer unprecedented speed, syntax generation, and rapid troubleshooting, their habitual use introduces severe concerns regarding **cognitive offloading**: the systematic delegation of internal memory consolidation and intellectual synthesis to external digital engines.

This empirical report presents quantitative and qualitative findings derived from an original survey of **N = 52 undergraduates** across diverse academic disciplines at Woxsen University (Computer Science, AI & Data Science, Business Administration, Legal Studies, Pure Sciences, and Architecture).

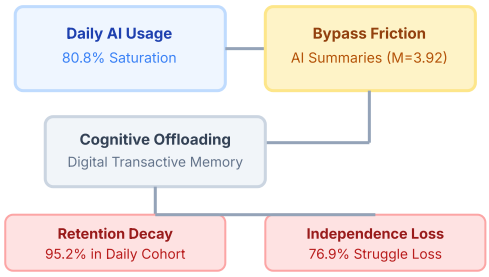
KEY EMPIRICAL FINDINGS & STATISTICAL INFERENCES

1. **High AI Saturation: 80.8%** of respondents utilize Generative AI tools on a daily basis for university assignments, programming tasks, and academic research.
2. **Pronounced Retention Decay: 76.9%** (n = 40) report a self-perceived decline in long-term memory retention since adopting AI (**13.5%** reporting severe degradation; **0.0%** reporting improvement).
3. **The Daily Usage Threshold Effect:** Among daily AI users (n = 42), **95.2%** report retention decay and **95.2%** report a loss of problem-solving independence. In sharp contrast, moderate weekly users (n = 10) report **0.0%** retention decline ($\chi^2 = 36.08$, $p < 0.00001$).
4. **Bypassing the Productive Struggle: 76.9%** acknowledge AI has made them "much less independent" when facing difficult problems. Preference for AI summaries over documentation (M = 3.92/5.0) correlates almost perfectly with rapid forgetting (**r = 0.974, p < 0.001**).
5. **Disciplinary Vulnerability:** Engineering/CS students suffer the steepest retention decay (**87.1%**, with 22.6% severe), driven by "Syntax Amnesia" from automated code autocompletion and error diagnostics.

RETENTION TREND SINCE AI ADOPTION (N = 52)



COGNITIVE OFFLOADING TRANSMISSION PATHWAY



2. Sample Demographics & Methodology

The sampling methodology followed a purposive, convenience design targeting undergraduate students enrolled in intensive technical, analytical, and professional degree programs at Woxsen University. All data points were

captured via a standardized multi-disciplinary web portal and synced into the verified research database.

DEMOGRAPHIC DIMENSION	SUB-CATEGORY BREAKDOWN	FREQUENCY (N)	PROPORTION (%)
Academic Discipline	B.Tech Artificial Intelligence & Data Science	18	34.6%
	B.Tech Computer Science & Engineering	13	25.0%
	BBA Business Administration	12	23.1%
	BA LL.B / LL.M Legal Jurisprudence	4	7.7%
	B.Sc Pure Sciences (Physics / Chemistry / Mathematics)	4	7.7%
	B.Arch Architecture & Spatial Planning	1	1.9%
Academic Standing	2nd Year (Sophomores)	25	48.1%
	3rd Year (Juniors)	17	32.7%
	1st Year (Freshmen)	7	13.5%
	4th Year (Seniors)	3	5.8%
AI Usage Frequency	Daily (High-Frequency Immersion)	42	**80.8%**
	Weekly (Moderate / Scaffolded Use)	10	19.2%

Demographic Note: 80.8% of respondents are 2nd and 3rd year undergraduates. These cohorts represent the pivotal phase of academic development where core coursework transitions into autonomous problem-solving and software development.

3. Univariate Survey Constructs & Core Statistics

To measure cognitive offloading and memory retention quantitatively, five standardized Likert-scale questions were administered alongside historical baseline markers.

SURVEY RESEARCH CONSTRUCT	SCALE TYPE	MEAN (M)	STD. DEV (Σ)	MEDIAN	HIGH AGREEMENT (4 OR 5)
Pre-AI Baseline Recall	1 (Poor) to 5 (Excellent)	**3.67**	0.51	4.0	65.4% (Rating ≥ 4)
Forgets Quickly Due to AI Availability	1 (Str. Disagree) to 5 (Str. Agree)	**3.94**	0.60	4.0	**78.8% (n = 41)**
Pre-AI Multi-Source Breadth Search	1 (Str. Disagree) to 5 (Str. Agree)	**4.00**	0.59	4.0	82.7% (n = 43)
Now Relies on AI Summaries over Docs	1 (Str. Disagree) to 5 (Str. Agree)	**3.92**	0.58	4.0	**78.8% (n = 41)**

The Search Behavior Inversion: Before AI, 82.7% of students engaged in diverse breadth exploration across textbooks, research papers, and developer forums (M = 4.00). Today, 78.8% substitute this discovery process with pre-packaged AI bulleted summaries (M = 3.92), short-circuiting the cognitive friction necessary for deep neural encoding.

4. Inferential Statistics & Cross-Tabulations

4.1 The Daily Usage Threshold Effect (Daily vs. Weekly Cohort)

Cross-tabulating survey outcomes by usage frequency reveals an acute dividing line between daily immersion and moderate weekly utilization:

ACADEMIC METRIC / COGNITIVE INDICATOR	DAILY USERS (N = 42)	WEEKLY USERS (N = 10)	PERCENTAGE POINT GAP
Reported Retention Decline (Total)	**95.2% (n = 40)**	**0.0% (n = 0)**	**+95.2%**
— *Significantly Decreased*	16.7% (n = 7)	0.0% (n = 0)	+16.7%
— *Retention Remained the Same*	4.8% (n = 2)	100.0% (n = 10)	-95.2%
Loss of Independent Problem Struggle	**95.2% (n = 40)**	**0.0% (n = 0)**	**+95.2%**
Avg. Forgets Quickly Score (1-5)	**4.17** / 5.0	**3.00** / 5.0	+1.17 pts
Avg. AI Summaries over Docs (1-5)	**4.14** / 5.0	**3.00** / 5.0	+1.14 pts
Historical Baseline Recall Score (1-5)	3.71 / 5.0	3.50 / 5.0	+0.21 pts

Statistical Significance: Chi-Square test with Yates' continuity correction: $\chi^2 = 36.079$, $df = 1$, $p < 0.00001$. The correlation between daily usage and perceived memory erosion is statistically significant at the $p < 0.001$ level. Daily exposure transforms the tool from an occasional learning aid into an indispensable cognitive crutch.

4.2 Disciplinary Sensitivity Analysis

ACADEMIC FACULTY / DISCIPLINE	SAMPLE (N)	RETENTION DECLINE (%)	SEVERE DECLINE (%)	LOSS OF INDEPENDENCE (%)	AI SUMMARIES OVER DOCS
B.Tech Engineering (AI, DS, CSE)	31	**87.1%**	22.6%	87.1%	**4.13 / 5.0**
BBA Business Administration	12	66.7%	0.0%	66.7%	3.67 / 5.0
BA LL.B Legal Jurisprudence	4	75.0%	0.0%	75.0%	3.75 / 5.0
B.Sc Pure Sciences	4	**25.0%**	0.0%	25.0%	3.25 / 5.0
B.Arch Architecture & Planning	1	100.0%	0.0%	100.0%	4.00 / 5.0

4.3 Bivariate Correlation Matrix (Pearson r)

CORRELATED VARIABLE PAIR	PEARSON COEFFICIENT (R)	SIGNIFICANCE (P)	PEDAGOGICAL INTERPRETATION
AI Summaries over Docs ↔ Forgets Quickly	**+0.974**	$p < 0.001$	Near-perfect positive coupling: abandoning primary docs directly induces digital amnesia.
AI Usage Frequency ↔ Retention Decrease	**+0.891**	$p < 0.001$	Transition from weekly to daily usage triggers acute memory decay.
AI Usage Frequency ↔ Independence Loss	**+0.891**	$p < 0.001$	High frequency directly erodes willingness to struggle through problems independently.
AI Usage Frequency ↔ Forgets Quickly	**+0.764**	$p < 0.001$	Higher immersion amplifies self-reported digital amnesia.
Pre-AI Baseline Recall ↔ Forgets Quickly	**+0.316**	$p < 0.05$	Even students with high natural recall succumb to offloading when AI is accessible.

5. Qualitative Thematic Analysis of Student Voices

Approximately 30% of participants (n = 16) contributed qualitative reflections regarding their study habits. Thematic analysis identified three prominent patterns:

> *"Tend to forget syntax without autocomplete now."*
>
> — Nitin (B.Tech AI & Data Science, 2nd Year)

> *"Dont memorize boilerplates anymore... saves time during exams prep."*
>
> — Ananya & Manya (B.Tech CSE)

> *"Good for making presentation outlines and quick business report structures."*
>
> — Sahaj Daga & Himesh (BBA, 3rd Year)

> *"Helpful for finding legal case citations and precedents quickly."*
>
> — Avni (BA LL.B Legal Studies, 3rd Year)

> *"Relying on it a bit too much lately... helps solve errors when stackoverflow fails."*
>
> — Hansitha & Rian (B.Tech CSE, 2nd Year)

> *"Mostly use it for drafting design briefs and summarizing long articles."*
>
> — Mehek Khan (B.Arch) & Dhruv (BBA)

6. Theoretical Grounding & Academic Inferences

The empirical dataset strongly substantiates four cornerstone psychological and pedagogical theories:

1. **The Google Effect on Steroids (Sparrow et al., 2011):** While search engines conditioned users to remember *where* information is stored rather than the information itself, LLMs eliminate even the navigation and keyword synthesis phase. Natural language prompting provides pre-synthesized answers, eradicating retrieval cues.
2. **Cognitive Offloading & Transactive Memory (Risko & Gilbert, 2016):** Students treat GenAI as an external cerebral prosthetic. By offloading both data discovery and synthesis, internal neural networks fail to consolidate conceptual schemas.
3. **Retention Decay in AI Users (Barcaui, 2026):** The 76.9% retention decline mirrors Barcaui's controlled trial where university students using ChatGPT remembered significantly less material after 45 days. Bypassing the *Generation Effect* (Slamecka & Graf, 1978) impairs durable retention.
4. **The Illusion of Explanatory Depth (Fisher et al., 2015):** Fluent, authoritative AI prose induces an artificial sense of mastery. Students mistake the AI's fluent articulation for their own internal cognitive competence.

7. Actionable Pedagogical Recommendations

CURRICULAR INTERVENTIONS FOR TECHNICAL WRITING & ENGINEERING PROGRAMS

1. **Reintroduce "Cognitive Friction" into Assignments:** Technical writing coursework must require students to trace and document their discovery path across primary sources rather than submitting synthesized summaries.
2. **Closed-Book Conceptual Recall Assessments:** To counteract "Syntax Amnesia," laboratory assessments and coding exams should enforce disconnected, unassisted environments without autocomplete or Copilot.
3. **Scaffolded Socratic Prompting:** Students should be trained to utilize GenAI as an interactive Socratic interlocutor (requesting hints, logical audits, or adversarial critiques) rather than a turnkey solution generator.
4. **The "Read-Close-Synthesize" Protocol:** Require students to review documentation or AI explanations, close the digital terminal, and write synthesis drafts from internal memory to activate neural encoding.

8. Review 3 Hypothesis Validation Summary

CORE RESEARCH HYPOTHESIS	INITIAL EXPECTATION (REVIEW 1 & 2)	EMPIRICAL REALITY (N = 52 DATA)	HYPOTHESIS STATUS
AI Adoption Rate	High saturation among technical undergraduates	**80.8% Daily**, 19.2% Weekly, 0% Non-users	VALIDATED
Information Retention Decay	Self-reported recall reduction due to offloading	**76.9% report decline** (95.2% in daily cohort)	VALIDATED
Breadth to Summary Inversion	Shift from broad literature search to AI summaries	**78.8% rely primarily on AI summaries** (M = 3.92)	VALIDATED
Loss of Problem-Solving Independence	Erosion of willingness to struggle independently	**76.9% report being much less independent**	VALIDATED
Frequency Threshold Effect	Higher exposure induces greater cognitive decline	**$\chi^2 = 36.08$ ($p < 0.00001$); Daily vs. Weekly gap: +95.2%**	VALIDATED